QND Computer Science Day 15

Mr. Schmidt

Today

- AI and Large Language Models
- AI Coding / Vibecoding
- Final Project

What AI tools have you used?

AI

- AI is not your friend
- AI is not alive
- It is a tool

How do they work?

- Collect a massive dataset
- "Tokenize" the data (turn it into a list of words or subwords)
- Train a model to predict the next token
- By continuing to predict the next token, they can generate convincing text

AI Coding

- Claude Code changed the game
- I manually write very little code

Why bother learning to code at all?

- Virtue in learning
- Knowing the limits
- The people do not want slop!

Your First AI Studio Project

- Rebuild the Platformer Game

Final Project Requirements

- Make something cool with AI Studio
- Keep it appropriate
- Be creative and have fun!
- The focus is *building something* not *playing something*
  - You'll have to use your app to iterate, but you should be adding features and improving it, not just playing a game
- On Friday, I'll show either your AI Studio app *or* your turtle project

Final Project Ideas

- Take your Adventure game and make a visual novel version
- Make a personal dashboard to track homework assignments, todos, events, etc.
- Build a quiz app about a topic you love
- Make a journaling app
- Make a website for a school club or sports team
- Try the "I'm feeling lucky" button and see what you get!

Sharing and Publishing

- Use the "Share" button in AI Studio to get a link
  - Limit to certain people or make it public
  - Runs within AI Studio
- To make it a standalone web app, click "Publish"
  - You get a URL that you can share with anyone and it will run without needing to log in to AI Studio
  - Requires a Google Cloud account and billing
  - $300 in free credits for new users goes a long way!
- Can add a link to your iPad / iPhone home screen for easy access!

Writing Prompts

❌ "Make me a cool app"
✅ "Build a quiz app about NBA basketball with 10 multiple choice questions. Show one question at a time, highlight the correct answer in green if I get it right or red if I get it wrong, track my score, and show my final score with a message at the end."

- Detail and specificity are key to getting good results